

SARAL GUPTA

B.Tech. Electrical Engineering

UG(II Year)

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Profiles: [Codeforces](#) | [CodeChef](#) | [GitHub](#) | [LinkedIn](#)

Indian Institute of
Technology
Roorkee



Area of Interest

Data Structures and Algorithms, Software Development, Machine Learning

Education

Year	Degree/Examination	Institution/Board	CGPA/Percentage
2026	B.Tech. 2nd Year	Indian Institute of Technology, Roorkee	8.617
2024	12th	Excellent Academy, Mandleshwar	86 %
2022	10th	Shri Kanwartara Public Higher Secondary School, Mandleshwar	93.2 %

Competitive Programming / Data Structures and Algorithms

Codeforces (saralgupta95): Maximum rating **1435**, Best global rank **2110** (*Div. 2 Round 1057*)

CodeChef (saralgupta95): Highest rating **1792** — Global Rank **44** in *Starters 211* (*Rated*)

Solved **90000+** problems across **Codeforces, LeetCode, and GeeksforGeeks**

Strong proficiency in **C++**, with solid understanding of **DSA** (arrays, recursion, sorting, binary search, trees, graphs, dynamic programming)

Projects

CNN Based House Price Prediction Model | IIT Roorkee

- Built a **CNN-based hybrid house price prediction model** using satellite images + tabular features ($R^2 \approx 0.86$).
- Implemented **custom multimodal training pipelines** for synchronized image and tabular data.
- Automated **satellite image acquisition via ArcGIS REST API**, enabling large-scale geospatial data collection for CNN training.
- Improved performance over tabular models through **deep spatial feature extraction** (CNNs).
- Applied **Grad-CAM visualizations** to interpret price-driving regions in satellite imagery.

Technologies: Python, TensorFlow/Keras, CNNs, Pandas, NumPy, Scikit-learn

Project Repository : <https://github.com/saralgupta09/House-Price-Prediction-Model>

Memory Allocation and CPU Cache Simulation in C | IIT Roorkee

- Designed and implemented a **C-based memory simulator** modeling heap allocation, cache hierarchy, and memory access behavior.
- Built a **custom dynamic memory allocator** (malloc/free) using an emulated sbrk interface.
- Simulated **multi-level CPU caches (L1/L2/L3)** with configurable block size, associativity, and replacement policies.
- Instrumented detailed **cache statistics (hits, misses, access patterns)** to analyze memory performance.

Technologies: C, Memory Allocation, Cache Architecture, Pointers, Low-level Systems Programming

Project Repository : <https://github.com/saralgupta09/memory-simulator>

Awards / Scholarships / Academic Achievements

JEE Advanced 2024: All India Rank 3564

JEE Main 2024 : 99.67 percentile, AIR 5345 among 14.5 lakh candidates

Academic Excellence (IIT Roorkee):

- Data Structures & Algorithms** — Grade 10 (Top 40 / 400 students(approx))
- Probability & Statistics** — Grade 10 (Top 140 / 1400 students(approx))
- Network Theory** — Grade 10 (Top 14 / 170 students)
- Linear Algebra & Vector Calculus** — Grade 9

Skills

Programming Languages: C++, Python, C, HTML, CSS

ML & Data Science: TensorFlow,Keras, Scikit-learn, Pandas, NumPy, Matplotlib

Systems & CS: Memory Allocation, Cache Architecture, Pointers, Low-level C,Low Level Design

Tools: Git, VS Code

Key Coursework

Data Structures & Algorithms, Probability & Statistics, Linear Algebra & Vector Calculus,Network Theory, Signals and Systems

Positions of Responsibility & Extra Curriculars

Branch Coordinator, Student Mentorship Programme (SMP), IIT Roorkee

- Acted as the primary liaison between seniors and first-year students of the branch; coordinated mentoring activities, addressed academic concerns, and ensured smooth information flow.

Member, Electrical Vertical, Team KNOx, IIT Roorkee

- Contributed to the electrical subsystem of the team by assisting in circuit design, component selection, and integration with mechanical and control modules.